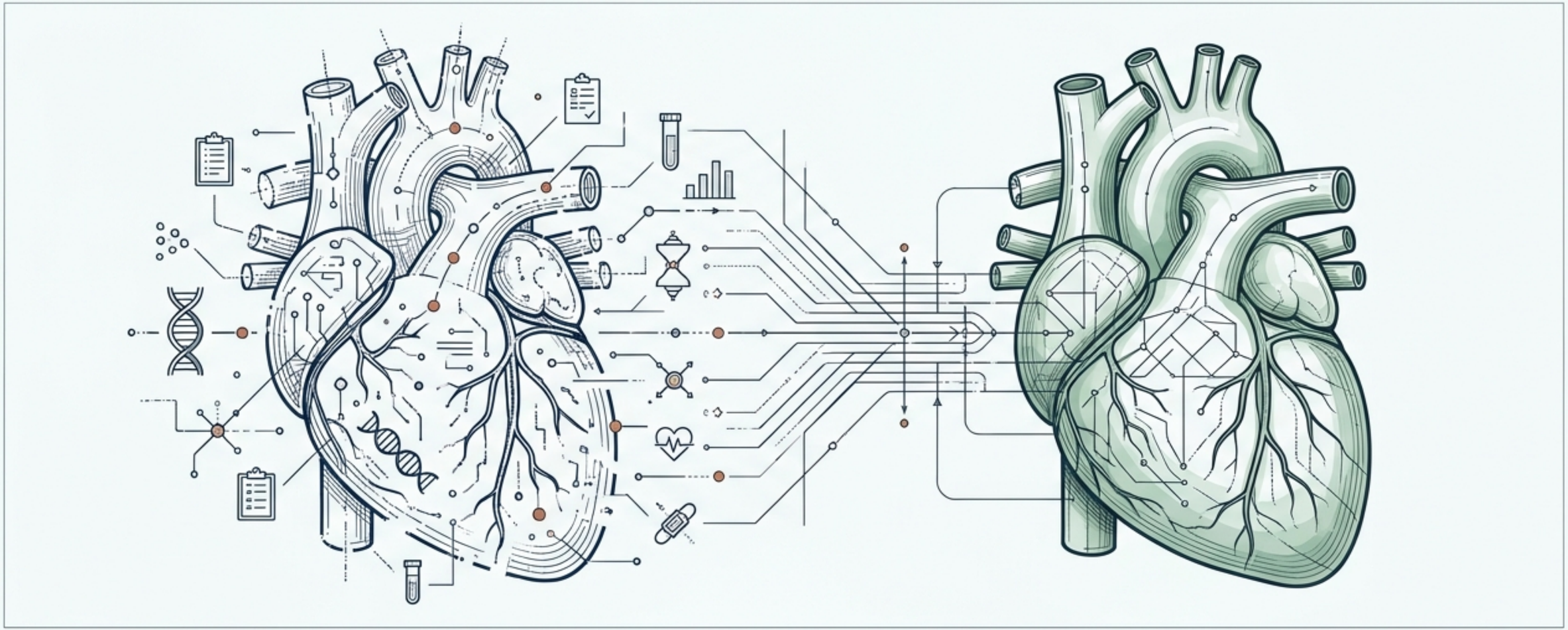
AI-Driven Personal Healthcare Digital Twin

From fragmented medical records to a continuously updated predictive patient model

Hackathon Technical Architecture Document



Gemini Notebook

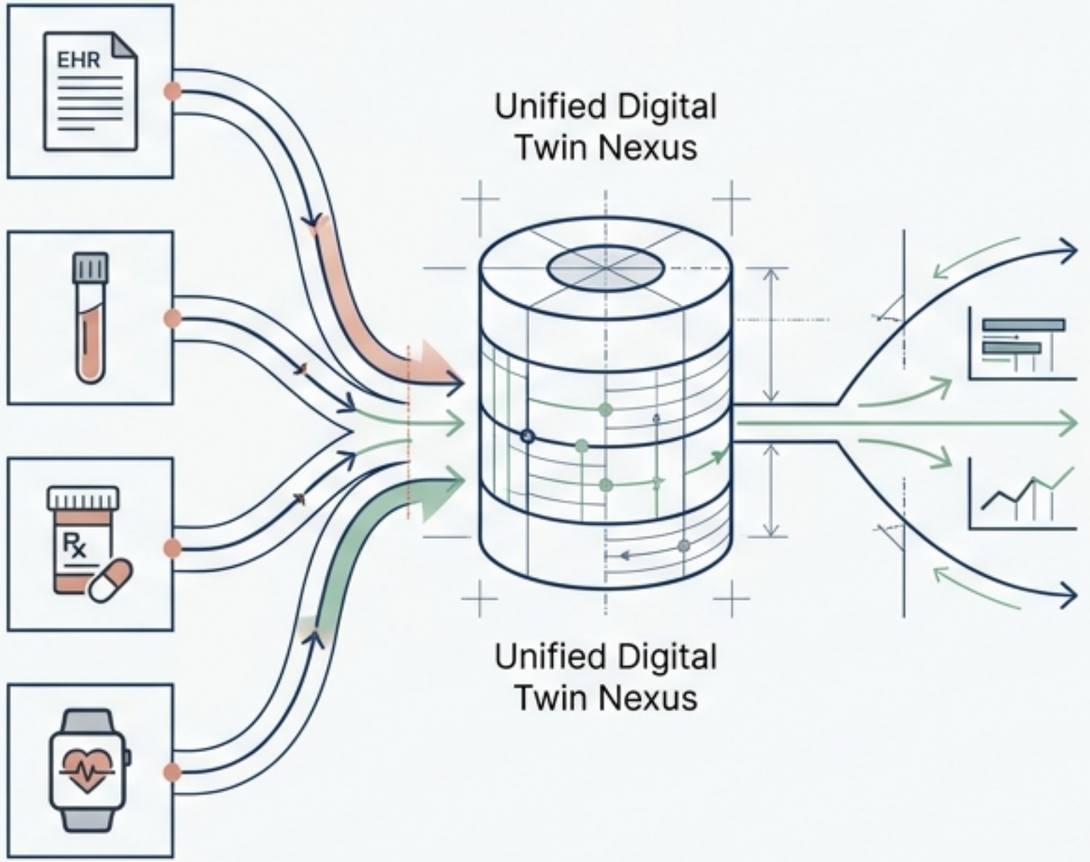
Traditional System

What Happened? (Retrospective & Siloed)



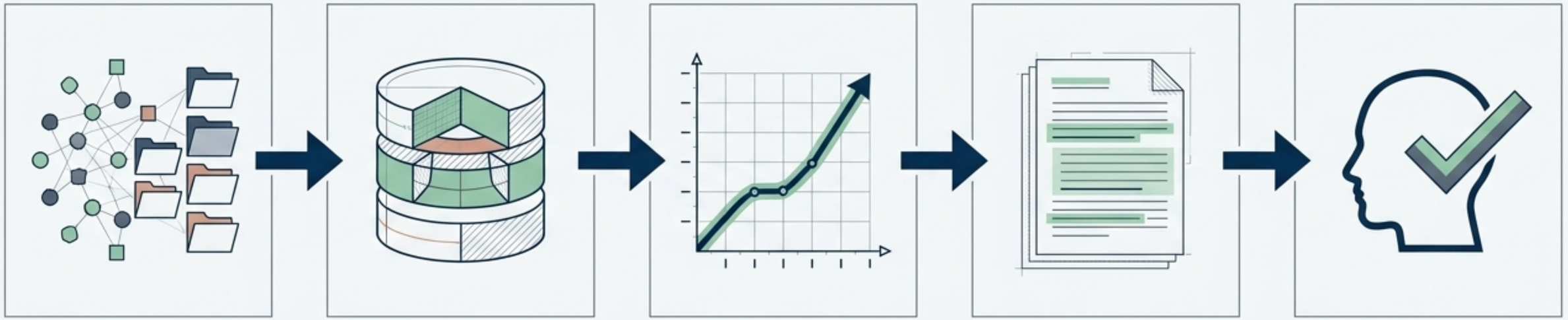
Digital Twin

What is happening, what may happen next, and why? (Dynamic & Predictive)



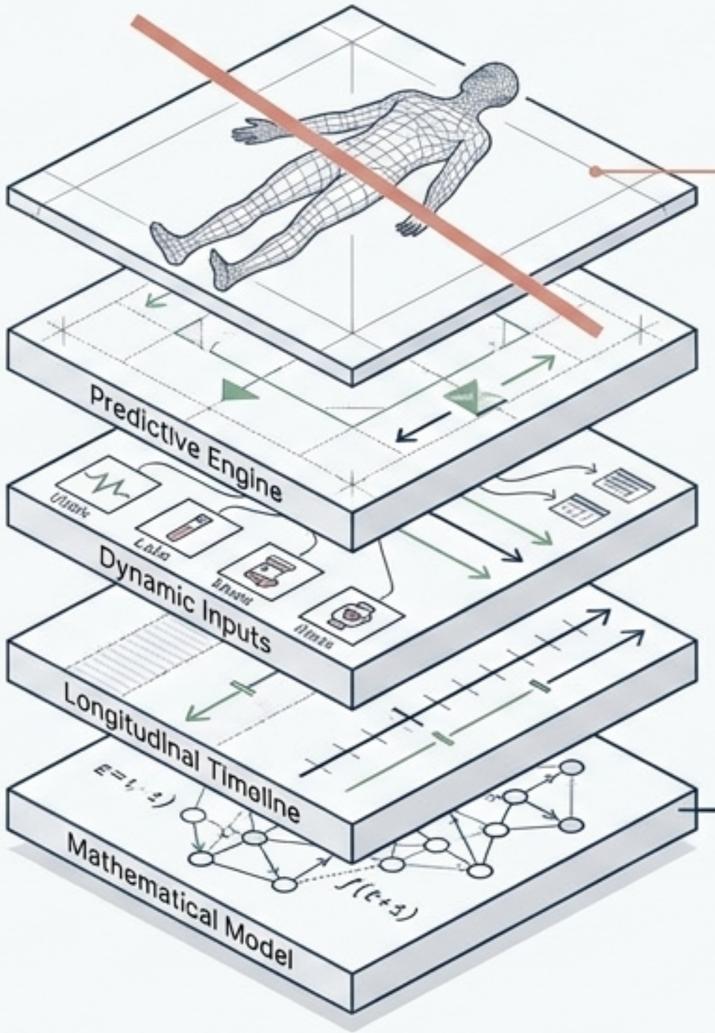
Gemini Notebook

Transforming raw medical artifacts into actionable clinical foresight



Gemini Notebook

A living computational node rather than a 3D visualization

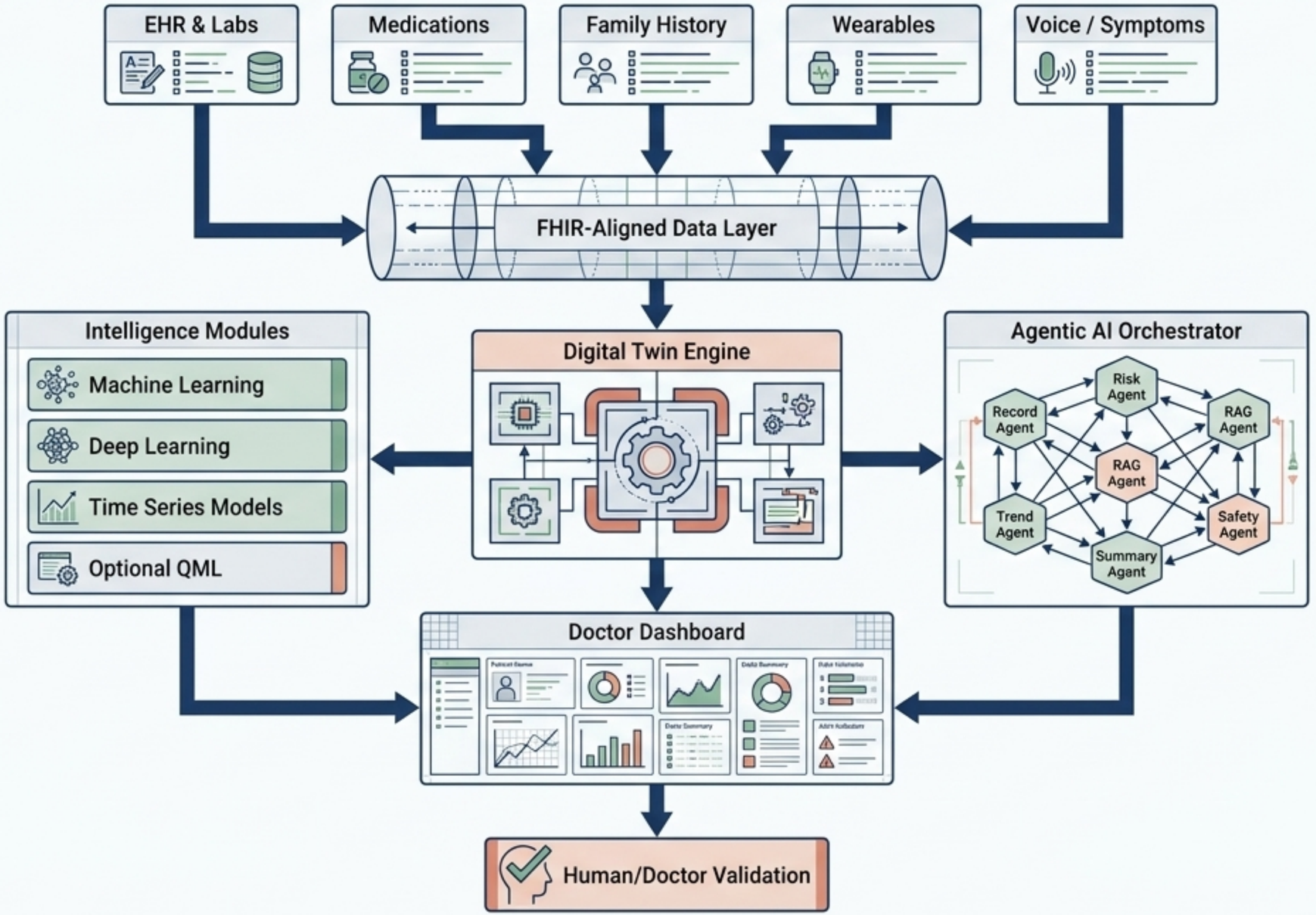


NOT a 3D visualization.

```
State(t+1) =
  f(Historical State
  + Current Vitals
  + Interventions)
```

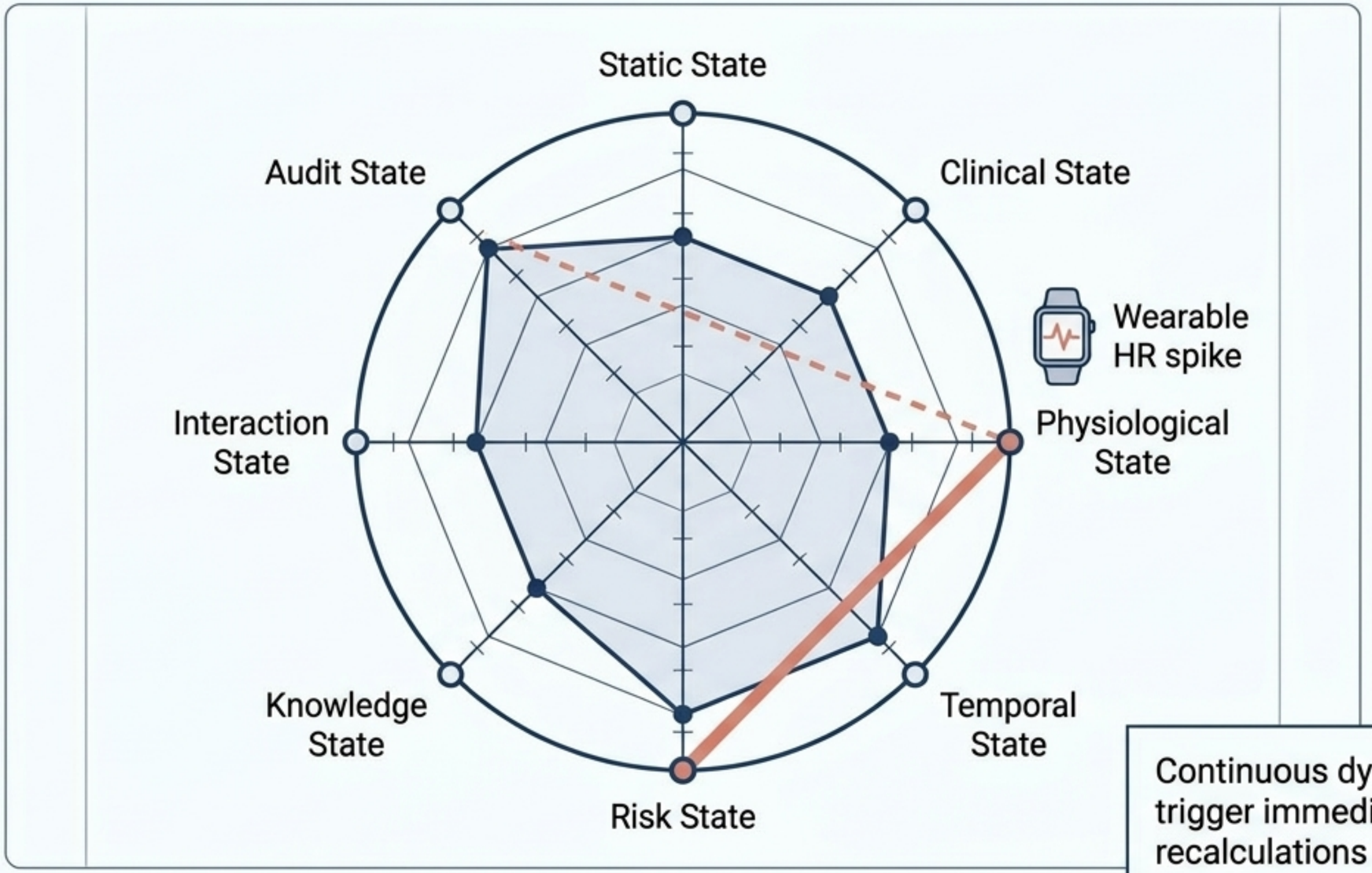
Gemini Notebook

The architectural core mapping data ingestion to clinical validation



Gemini Notebook

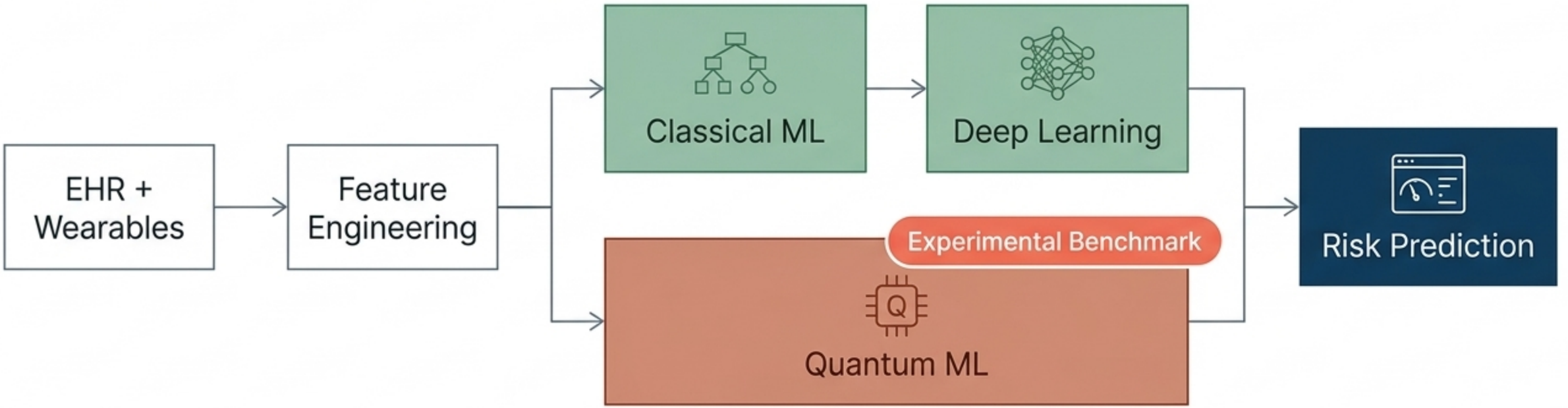
The interconnected dimensions of the Digital Twin state



Continuous dynamic updates trigger immediate state recalculations across the twin.

Gemini Notebook

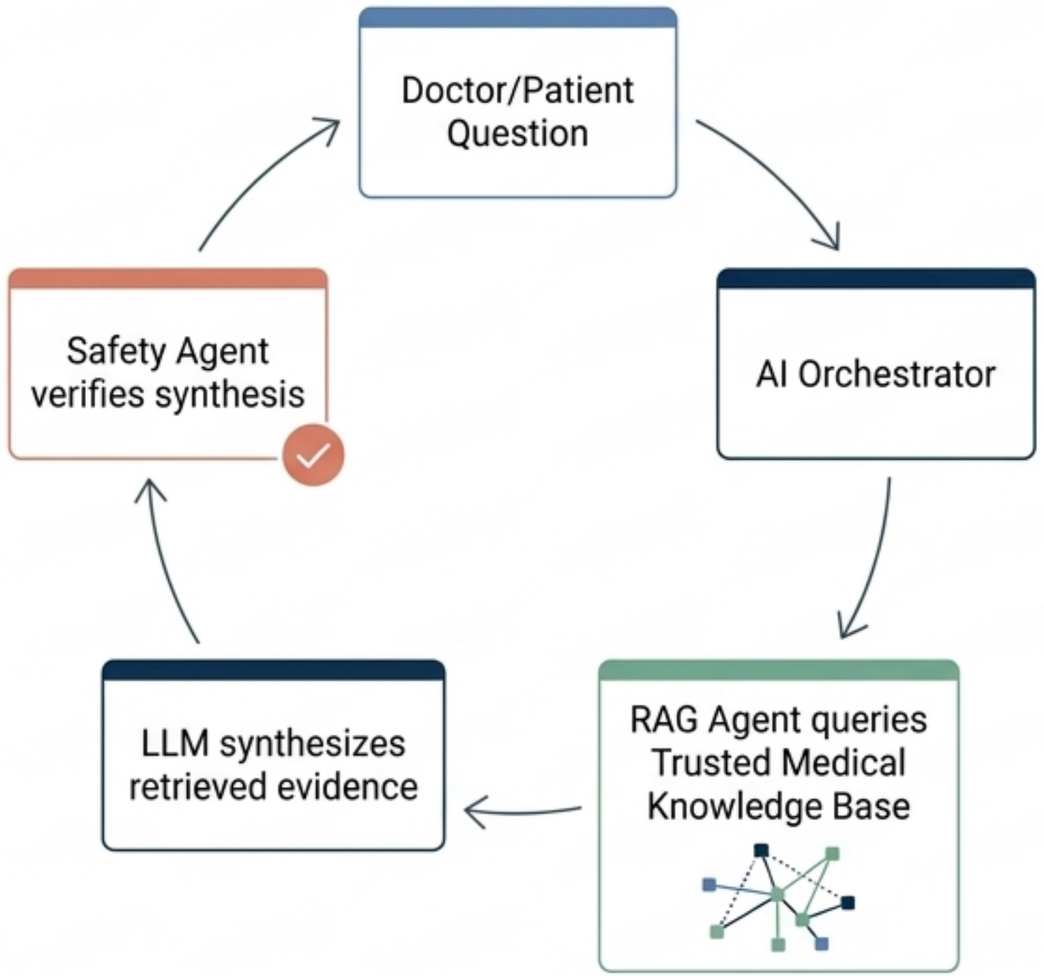
Benchmarking predictive modalities across high-dimensional feature spaces



Hypothesis: Testing QML for high-dimensional temporal feature spaces. We do not presume automatic superiority over classical Deep Learning, but seek empirical latency-to-accuracy comparisons.

Gemini Notebook

Multi-agent orchestration grounded in verified medical evidence



AI-Assisted Clinical Response

Based on recent clinical data, for adult patients with type 2 diabetes, metformin remains the preferred initial pharmacologic agent[1]. If the A1C target is not achieved after approximately 3 months, a second oral agent or injectable therapy should be considered[1].

However, in patients with established atherosclerotic cardiovascular disease or kidney disease, a sodium-glucose cotransporter 2 (SGLT2) inhibitor or a glucagon-like peptide 1 (GLP-1) receptor agonist with demonstrated cardiovascular benefit is recommended, independent of baseline A1C[1].

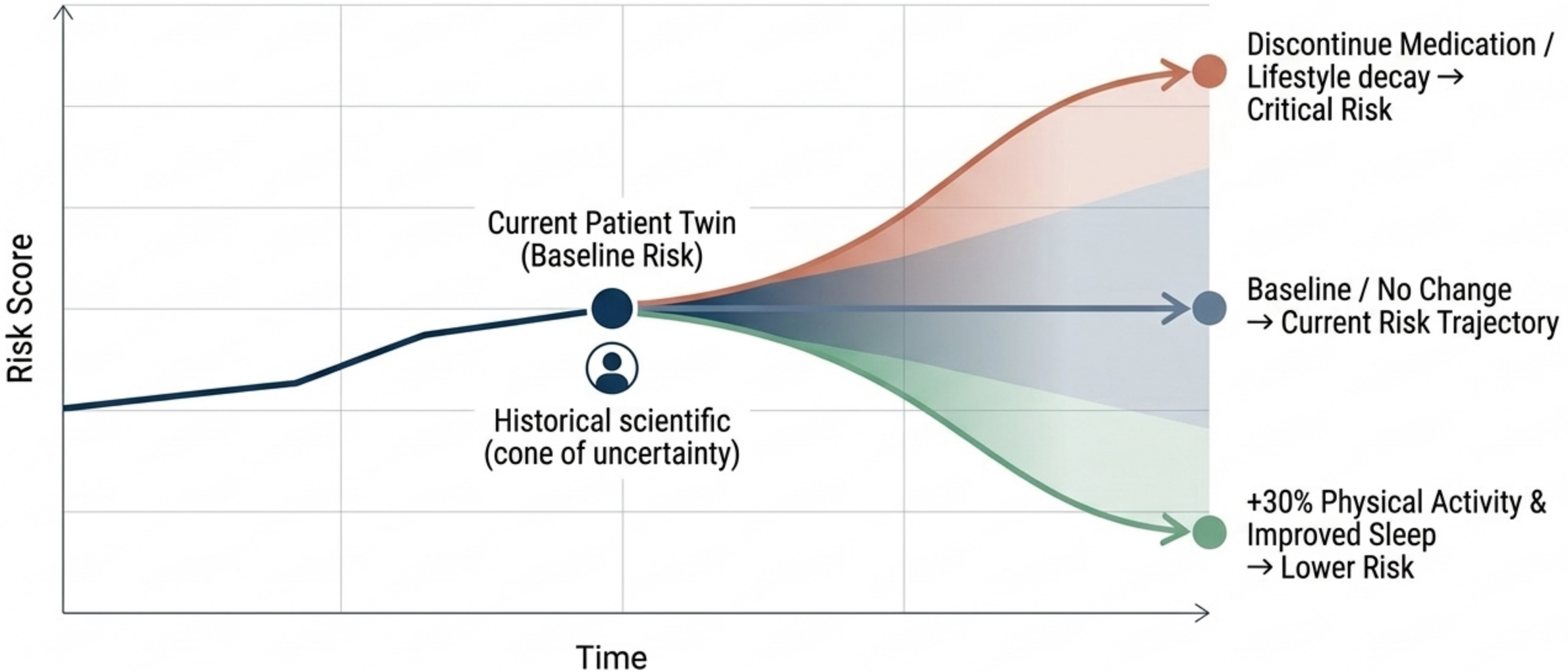
Patient EHR data indicates a creatinine level of 1.2 mg/dL on 08/12[2], suggesting stable renal function.

References

- [1] ADA Guidelines 2023
- [2] Patient EHR Lab 08/12

Gemini Notebook

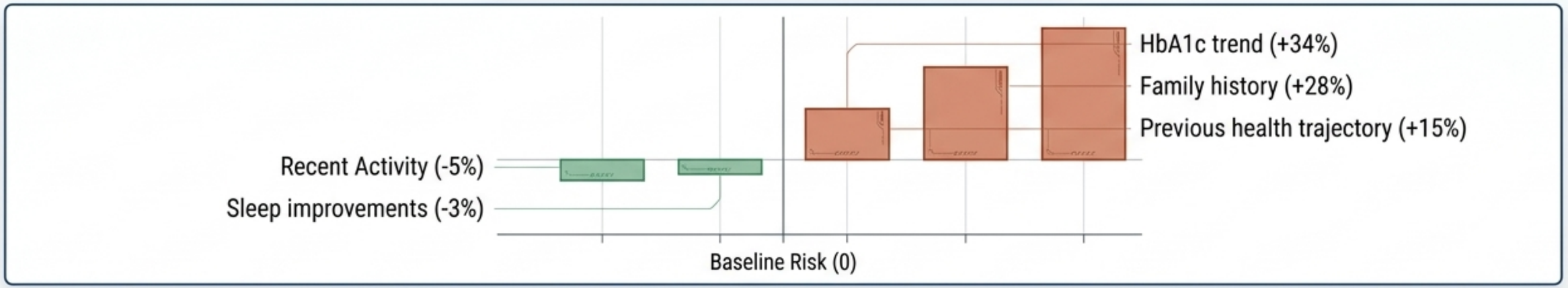
Simulating interventions to forecast diverging patient outcomes



Gemini Notebook

Transparent decision support requiring clinician oversight





[Accept AI Recommendation] ✓ **[Modify Parameters]** **[Reject / Override]** ×

AI Prediction → Doctor Review → Action. Supporting clinicians rather than replacing them.

Gemini Notebook

Bridging continuous patient inputs with high-density clinical review

Patient App: Continuous Input & Interaction

Multilingual Voice Interface



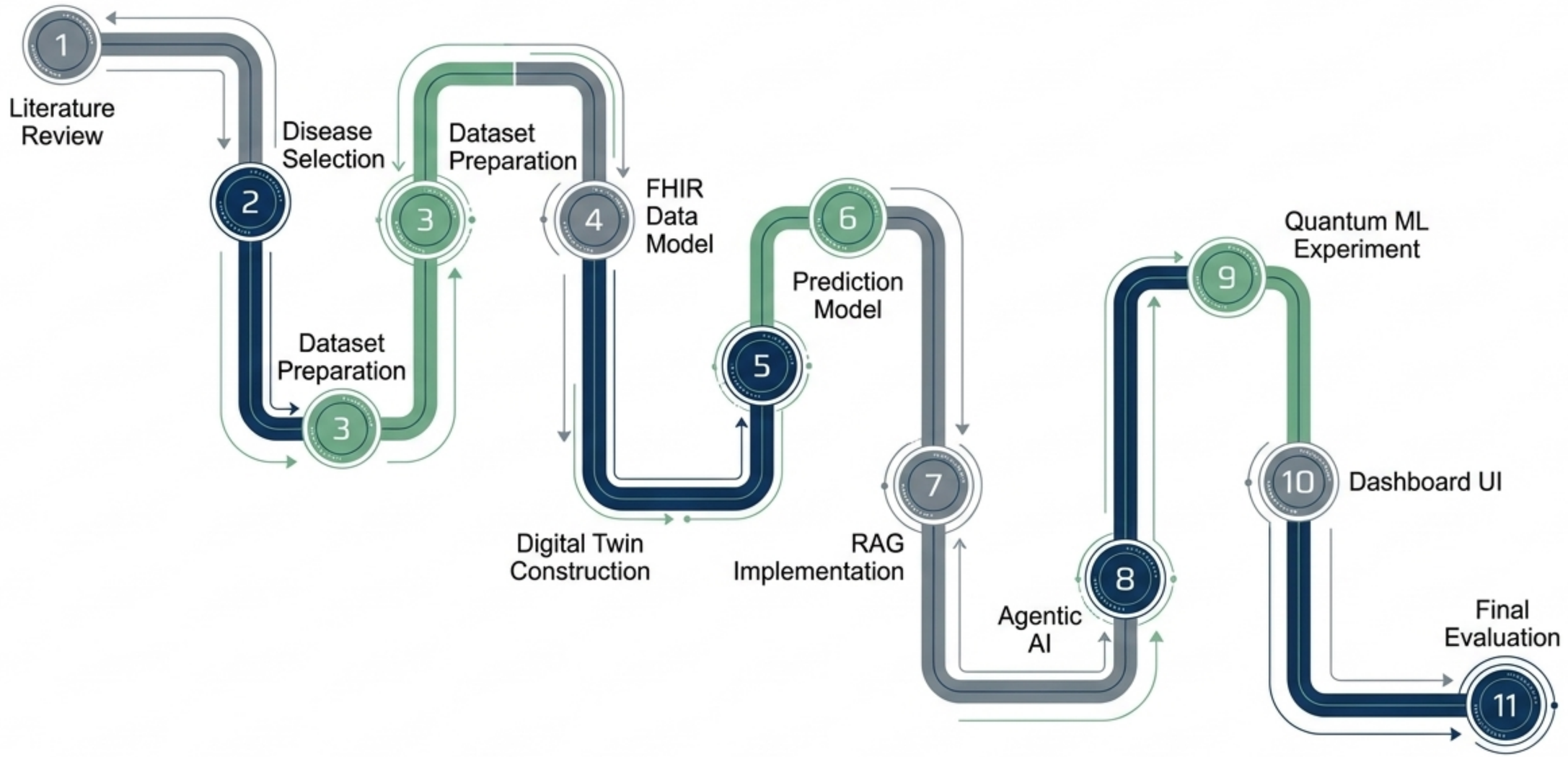
Doctor Dashboard: High-Density Twin Interaction



Continuous Patient Timeline

Gemini Notebook

Rigorous construction from literature to empirical evaluation



Gemini Notebook

Evaluating models and workflows across clinical benchmarks

Predictive Models							
	AUROC	AUPRC	Precision	Recall	F1	Calibration	MAE/RMSE
Classical ML	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX
Deep Learning	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX
Quantum ML	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX	0.XX

Generative Workflows	Groundedness (%)	Citation Coverage	Latency
Single LLM	0.XX	0.XX	0.XX
RAG-grounded LLM	0.XX	0.XX	0.XX
Agentic Workflow	0.XX	0.XX	0.XX

Gemini Notebook

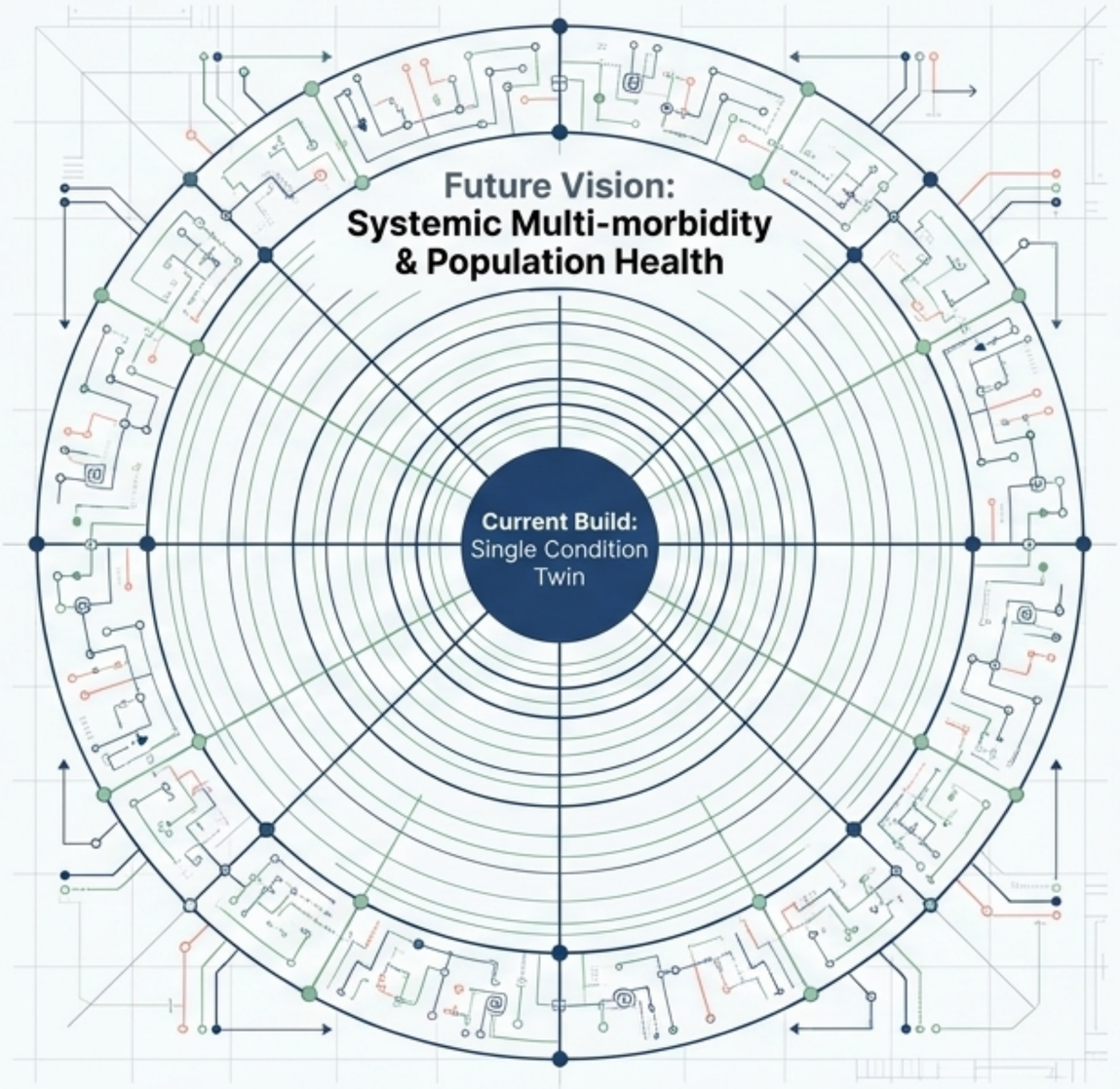
From single-disease prediction to systemic population health

Current Research Gaps

Integration of continuous hardware sensor streams.

Federated learning for enhanced data privacy.

Expanding the FHIR ontology for deeper physiological states.



Fusing Agentic orchestration, RAG, and time-series forecasting to evolve the medical record from a static history into a living, predictive engine.

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